SI 140A-01 Probability & Statistics for EECS, Fall 2026
Homework 6

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

(a) For $X \sim \text{Pois}(\lambda)$ and any function g, derive the relationship between $E[Xg(X)]$ and $E[g(X+1)]$. This is called the Stein-Chen identity for the Poisson.

(b) Find the moment $E\left(X^4\right)$ for $X \sim \text{Pois}(\lambda)$ by using the identity from (a) with the fact that X has mean λ and variance λ.

(a) $X \sim \text{Pois}(\lambda)$, since $k=0$, $k g(k) \Pr(X=k) = 0$,

$$E[Xg(X)] = \sum_{k=1}^{\infty} k g(k) e^{-\lambda} \frac{\lambda^k}{k!} = \lambda \sum_{k=1}^{\infty} g(k) e^{-\lambda} \frac{\lambda^{k-1}}{(k-1)!} = \lambda \sum_{k=0}^{\infty} g(k+1) e^{-\lambda} \frac{\lambda^k}{k!}$$

$$= \lambda E[g(X+1)]$$

$$\Rightarrow E[Xg(X)] = \lambda E[g(X+1)]$$

(b) $E[X] = \lambda$, $\text{Var}(X) = \lambda$.

$$E[X^2] = \text{Var}(X) + (E[X])^2 = \lambda + \lambda^2.$$

let $g(X) = X^2$, $E[X^3] = \lambda E[(X+1)^2] = \lambda E[X^2 + 2X + 1] = \lambda(E[X^2] + 2E[X] + 1) = \lambda^3 + 3\lambda^2 + \lambda$.

let $g(X) = X^3$, $E[X^4] = \lambda E[(X+1)^3] = \lambda(E[X^3] + 3E[X^2] + 3E[X] + 1)$

$$E[X^4] = \lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda$$

Problem 2

Alice flips a fair coin n times and Bob flips another fair coin $n+1$ times, resulting in independent $X \sim \text{Bin}(n, \frac{1}{2})$ and $Y \sim \text{Bin}(n+1, \frac{1}{2})$.
(a) Let $V = \min(X, Y)$ be the smaller of X and Y, and let $W = \max(X, Y)$ be the larger of X and Y. (If $X = Y$, then $V = W = X = Y$.) Find $E(V) + E(W)$.
(b) Show that $P(X < Y) = P(n - X < n + 1 - Y)$.
(c) Compute $P(X < Y)$. Hint: Use (b) and the fact that X and Y are integer-valued.

(a) $X+Y = V+W$,

$E(V)+E(W) = E(X)+E(Y) = \frac{n}{2} + \frac{n+1}{2} = n+\frac{1}{2}$.

(b) $n-X : \bar{X}$, $n+1-Y : \bar{Y}$

the swap sequence has the same probability.

so head counts (X, Y) become $(n-X, n+1-Y)$

Therefore, $P(X<Y) = P(n-X < n+1-Y)$

(c) $n-X < n+1-Y \Rightarrow Y-X < 1$. X, Y are integer-valued $\Rightarrow Y-X \le 0$, $Y \le X$.

so $P(X<Y) = P(Y \le X)$, and $P(X<Y) + P(Y \le X) = 1$, the two events are complementary.

$P(X<Y) = \frac{1}{2}$.

Problem 3

Consider two probability distributions $P = (0.75, 0.25)$ and $Q = (0.5, 0.5)$ over a binary random variable $X \in \{0, 1\}$. All logarithms are base 2.

(a) **Basic Calculations (5 points)**

Compute the total variation distance and KL divergence:

$$d_{\mathrm{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|,$$

$$D_{\mathrm{KL}}(P\|Q) = \sum_x P(x) \log_2 \frac{P(x)}{Q(x)}.$$

Is KL divergence symmetric? Justify your answer.

(b) **Connection to Information Theory (5 points)**

i) Compute the entropy $H(P)$ and cross-entropy $H(P, Q)$, which are defined as

$$H(P) = -\sum_x P(x) \log_2 P(x),$$

$$H(P, Q) = -\sum_x P(x) \log_2 Q(x).$$

ii) Show that

$$H(P, Q) - H(P) = D_{\mathrm{KL}}(P\|Q).$$

Briefly explain the significance of this identity in lossless data compression.

(c) **Connection to Machine Learning (5 points)**

Suppose P represents the true label distribution in a binary classification task. Two models predict $Q = (0.5, 0.5)$ and $R = (0.95, 0.05)$, respectively.

Compute $D_{\mathrm{KL}}(P\|Q)$ and $D_{\mathrm{KL}}(P\|R)$. Which model has a smaller expected cross-entropy loss? Explain your answer.

(a) $d_{TV}(P,Q) = \frac{1}{2}\sum_x |P(x) - Q(x)| = \frac{1}{2}(|0.75-0.5| + |0.25-0.5|) = 0.25$

$D_{KL}(P\|Q) = \sum_x P(x)\cdot \log_2 \frac{P(x)}{Q(x)} = 0.75 \log_2 \frac{0.75}{0.5} + 0.25 \log_2 \frac{0.25}{0.5} \approx 0.1887$ bits

the reverse direction $D_{KL}(Q\|P) = 0.5 \log_2 \frac{0.5}{0.75} + 0.5 \log_2 (\frac{0.5}{0.25}) \approx 0.2075$ bits, KL divergence is not symmetric.

(b) $H(P) = -\sum_x P(x) \log_2 P(x) = -0.75 \log_2 0.75 - 0.25 \log_2 0.25 \approx 0.8113$ bits

$H(P,Q) = -\sum_x P(x) \log_2 Q(x) = -0.75 \log_2 0.5 - 0.25 \log_2 0.5 = 1$ bit

$H(P,Q) - H(P) = -\sum_x P(x) \log_2 Q(x) + \sum_x P(x) \log_2 P(x) = \sum_x P(x) \cdot \log_2 \frac{P(x)}{Q(x)} = D_{KL}(P\|Q)$, in lossless data compression, $D_{KL}(P\|Q)$ is the extra ideal average number of bits per symbol incurred by coding for Q when the source follows P, compared with coding for P.

(c) $D_{KL}(P\|Q) \approx 0.1887$ bits.

$D_{KL}(P\|R) = 0.75 \log_2 \frac{0.75}{0.95} + 0.25 \log_2 \frac{0.25}{0.05} \approx 0.3247$ bits.

$H(P,M) = H(P) + D_{KL}(P\|M)$, since $H(P,Q) = 1$ bit and $H(P,R) \approx 1.136$ bits,

the model predicting Q has the smaller expected cross-entropy loss.